STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



OPERATIONS

SIX SIGMA AND VARIATION

Where Six Sigma came from, what 3.4 defects per million really means, Shewhart's two kinds of variation, whether Six Sigma pays, and a card for DMAIC.

3.4 defects per million opportunities is the Six Sigma level, a figure that already assumes the process drifts by 1.5 sigma.

MOTOROLA, 1986

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NO. 25 · OPERATIONS

In this issue

Every process varies: the same task takes eight minutes one day and twelve the next. Some of that variation belongs to the process, some has a cause you can find. These pages show how to tell the two apart, and look at Six Sigma, the programme built to reduce variation.

Motorola dates Six Sigma to 1986. Its 3.4 defects per million opportunities assumes that a process drifts by 1.5 sigma over time. The idea behind it is older: in 1924 Walter Shewhart proposed the control chart to separate two sources of variation, and Deming estimated that 94% of troubles belong to the system. Two studies of 200 and 84 firms found that adopting Six Sigma went with better performance.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editor interprets rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

3.4 per million

Motorola dates the invention of Six Sigma to 1986. The name became a promise of near-perfect quality, and the arithmetic behind it rests on an assumption.

Six Sigma sets out to reduce the variation in processes, because that is where defects come from. A process at six sigma gives 3.4 defects per million opportunities: not per million products, since one product can fail in many places.

3.4
defects per million opportunities, with the 1.5 sigma shift

≈0.002
per million at ±6 sigma without the shift

The 3.4 rests on a convention. Over time, the mean of a process is assumed to drift by up to 1.5 sigma, so a six-sigma process behaves in the long run like a 4.5-sigma one. The shift is an assumption rather than a measurement.

OUR READING

Six Sigma is less a number than a habit: measuring variation before arguing about it.

Sources: Motorola Solutions, 2026; iSixSigma, 2026

THE NUMBERS

From three to six sigma

With the 1.5 sigma convention, each step up the scale cuts the defects per million opportunities sharply.

DEFECTS PER MILLION OPPORTUNITIES, BY SIGMA LEVEL, WITH THE 1.5 SIGMA SHIFT



93.32%
of opportunities without a defect at three sigma

99.99966%
at six sigma

OUR READING

The table is arithmetic on a convention; no factory measured it. What it shows is how far apart three and six sigma lie.

Source: iSixSigma, 2026

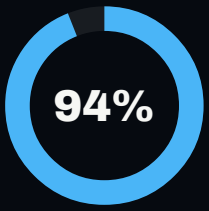
Calculated with the 1.5 sigma shift; some tables round three sigma to 66,810 or 66,800.

THE MODEL

Two kinds of variation

In an internal memo of 16 May 1924, Walter Shewhart of Western Electric proposed the control chart to his superiors. In it he separated two sources of variation.

Common causes	Special causes
Part of the process itself	Appear now and then
Reduced only by changing the process	Can be found and removed



of troubles and possibilities for improvement belong to the system, by Deming's estimate from experience; 6% are special.

OUR READING

Blaming a person for what the system produces changes nothing. Treating a special cause as normal lets it come back.

Sources: ASQ, Honorary Members, 2026 (via Excerpts, through web search; Wikipedia); W. Edwards Deming, MIT Center for Advanced Engineering Study, 1986 (via The W. Edwards Deming Institute)

Shewhart called them chance and assignable causes; Deming named them common and special. The 94% is Deming's estimate; it was never measured.

More in the essay A good SOP is not a document

WHAT THE RESEARCH SAYS

Does Six Sigma pay?

Two studies in the Journal of Operations Management in 2012 compared firms that adopted Six Sigma with similar firms that did not.

01	**200 firms**	Morgan Swink and Brian Jacobs matched each firm with control firms of similar return on assets before adoption, industry and size. Six Sigma had a positive effect on return on assets, mainly through significant cuts in indirect costs.
02	**84 firms, ten years**	Scott Shafer and Sara Moeller followed firms from three years before adoption to six years after, against control groups. Overall, adopting Six Sigma went with better firm performance.

OUR READING

Both studies look at firms that announced adoption. They show what tended to follow, which is no guarantee.

Sources: Morgan Swink & Brian W. Jacobs, Journal of Operations Management, 2012 (via Abstract); Scott M. Shafer & Sara B. Moeller, Journal of Operations Management, 2012 (via Abstract)

Associations rather than full proof of cause; we could not see the effect sizes of the second study.

HOW IT IS MEASURED

Reading a control chart

A control chart plots one measure over time, with limits three standard deviations from the mean. In a stable process a point falls outside them with a probability of 0.27%: about one false alarm in 370 points.

1. **Plot one measure in time order**
   Per day, shift or batch, always counted the same way.
2. **Draw the mean and the limits**
   Three standard deviations above and below the mean.
3. **Act on signals**
   A point outside the limits points to a special cause: look for it.
4. **Improve the process for the rest**
   Inside the limits, work on the process as a whole rather than on single points.

HYPOTHETICAL EXAMPLE, MINUTES TO LOAD A TRUCK

Mean	42 minutes
Limits	30 to 54 minutes
Day 17	61 minutes: outside, a forklift was down

Only day 17 calls for a search for a cause; the rest is the process. The numbers are made up.

Source: NIST & SEMATECH, 2026

The limits and the false-alarm rate follow NIST; the steps and the example are the editors'.

Open the tool: Sigma & Control Chart stivencatalyst.com/tools/sigma-control-chart/

TOOL OF THE ISSUE

The DMAIC card

DMAIC is ASQ's structured way to improve an existing process that falls short of its standard or of what the customer expects. One card per project.

01 Define the problem, the customer and the goal, on one page

02 Measure the baseline, and whether the measurement itself can be trusted

03 Analyze the main causes, shown with data

04 Improve the change, tested before it is rolled out

05 Control the control plan and the chart that keeps the gain

06 For a new process DMADV instead: define, measure, analyze, design, verify

Source: ASQ, 2026 (via Excerpts, through web search)

The five phases follow ASQ; the card is the editors' own.

Open the tool: Six Sigma in the Warehouse stivencatalyst.com/tools/six-sigma-dmaic/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Timeline
Motorola Solutions, 2026
https://www.motorolasolutions.com/en_xl/about/history/timeline.html

Sigma Performance Levels: One to Six Sigma
iSixSigma, 2026
https://isixsigma.com/basics/sigma-performance-levels-one-six-sigma

Walter A. Shewhart
ASQ, Honorary Members, 2026 · via Excerpts, through web search; Wikipedia
https://asq.org/about-asq/honorary-members/shewhart

Out of the Crisis
W. Edwards Deming, MIT Center for Advanced Engineering Study, 1986 · via The W. Edwards Deming Institute

Six Sigma adoption: Operating performance impacts and contextual drivers of success
Morgan Swink & Brian W. Jacobs, Journal of Operations Management, 2012 · via Abstract
https://doi.org/10.1016/j.jom.2012.05.001

The effects of Six Sigma on corporate performance: An empirical investigation
Scott M. Shafer & Sara B. Moeller, Journal of Operations Management, 2012 · via Abstract
https://doi.org/10.1016/j.jom.2012.10.002

NIST/SEMATECH e-Handbook of Statistical Methods: Cusum Average Run Length
NIST & SEMATECH, 2026
https://www.itl.nist.gov/div898/handbook/pmc/section3/pmc3231.htm

DMAIC Process: Define, Measure, Analyze, Improve, Control
ASQ, 2026 · via Excerpts, through web search
https://asq.org/quality-resources/dmaic

CORRECTION · 12 OCTOBER 2026

The history of Six Sigma now gives only what Motorola's own timeline says, that it was invented in 1986; the issue first also named Bill Smith, Bob Galvin and a launch in 1987, which the issue's sources do not give.

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what it measures, in the full text where it could be opened. A source read only in a summary, an abstract or excerpts is marked “via”. A figure not confirmed in what was read is marked so on its page, or left out. The editor's interpretation is marked “Our reading”.



MANAGEMENT REVIEW

Management without theatre.

Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html

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Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html